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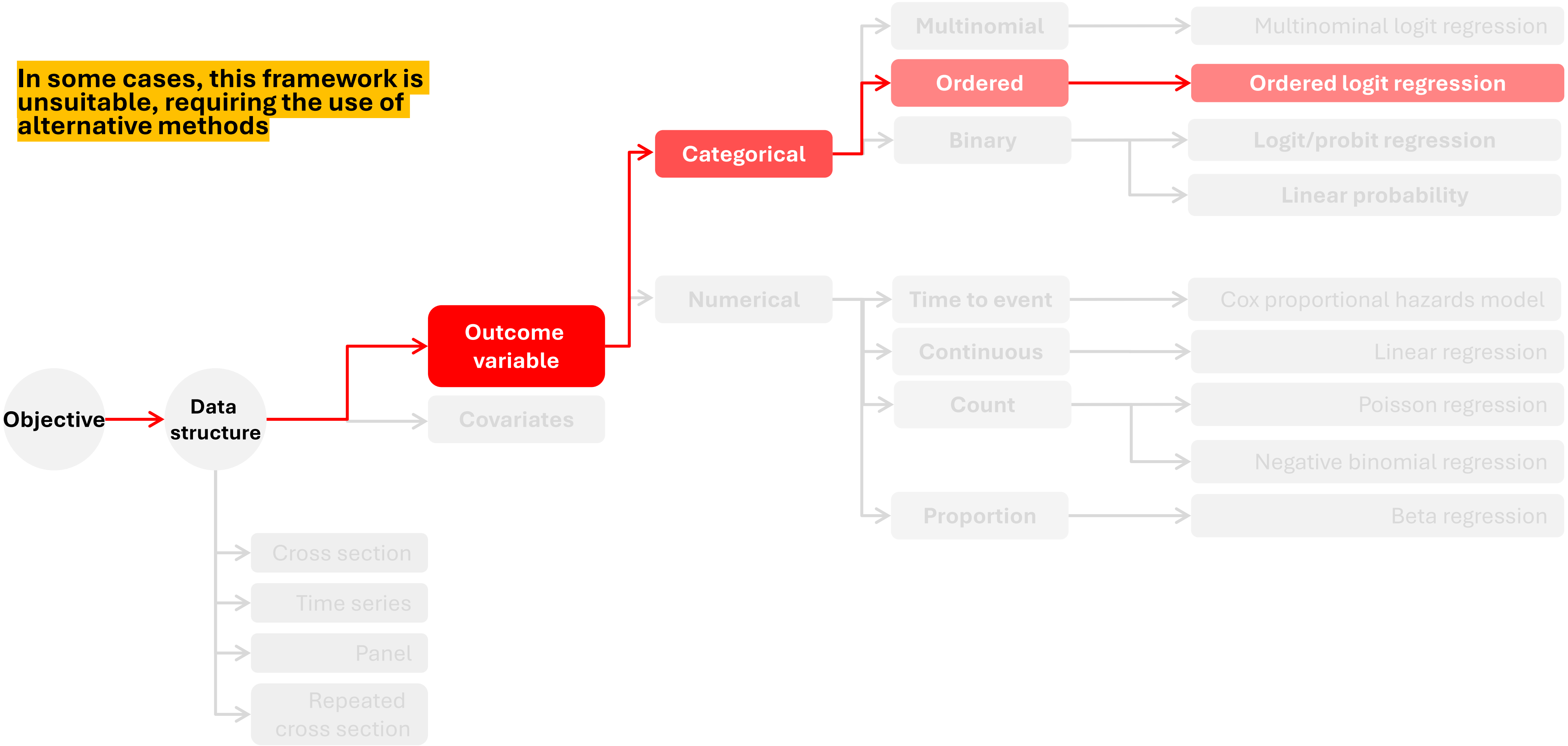
Universidad Externado de Colombia

Multinomial models

The ordered logit model

Data types and model selection

In some cases, this framework is unsuitable, requiring the use of alternative methods



Categorical dependent variables with multiple categories

So far, we have studied situations where **the dependent variable is either continuous or binary**.

- **Continuous dependent variables**
  Simple linear regression approaches
- **Binary dependent variables**
  Linear probability models and logistic (logit) models.

In the case of logit models, we looked at scenarios where the dependent variable takes two values:

- Employed (Y = 1) or unemployed (Y = 0)
- Participation in elections (Y = 1) or non-participation (Y = 0)

In many cases **the variable of interest can take more than two values**:

1. Nominal categorical dependent variables

$$\text{Region}(x) = \begin{cases} \text{Caribe} & \text{si } x \in \{\text{ATL}, \text{BOL}, \text{COR}, \text{LAG}, \text{MAG}, \text{SUC}, \text{SAP}\} \\ \text{Andina} & \text{si } x \in \{\text{ANT}, \text{BOY}, \text{CUN}, \text{NSA}, \text{SAN}, \text{TOL}, \text{CAL}, \text{RIS}, \text{QUI}, \text{HUI}\} \\ \text{Pacífica} & \text{si } x \in \{\text{CHO}, \text{VAC}, \text{CAU}, \text{NAR}\} \\ \text{Orinoquía} & \text{si } x \in \{\text{ARA}, \text{CAS}, \text{MET}, \text{VIC}\} \\ \text{Amazonía} & \text{si } x \in \{\text{AMA}, \text{CAQ}, \text{GUV}, \text{GUA}, \text{PUT}, \text{VAU}\} \end{cases}$$

2. Ordinal categorical dependent variables

$$\text{Satisfaction}(x) = \begin{cases} \text{Low} & \text{si } x = 1 \\ \text{Mid} & \text{si } x = 2 \\ \text{High} & \text{si } x = 3 \end{cases} \quad \text{where low} < \text{Mid} < \text{High}$$

Ordinal regression

1. Treat the dependent variable as if it were continuous

In this case, linear regression methods (OLS) can be used. **This approximation depends on the number of categories in the dependent variable**.

2. Ignore the ordinality of the dependent variable and treat it as nominal

Here, multinomial logistic models can be used.
In doing so:

- ✖ It discards part of the information available in the ordering
- ✖ It is possible that more parameters than necessary are estimated.
- ✖ It increases the risk of obtaining non-significant statistical results.

3. Treat the dependent variable as ordinal

We assume that the variable is measured on an ordinal scale, even if the scale is a rough measurement of an **underlying interval or ratio scale**.

For example, the categories:
- High
- Medium
- Low

Could be seen as approximations of socioeconomic status or intelligence.

Ordinal regression

Proportional Odds models

1. We observe an ordinal variable Y

2. Y, is a function of another variable , Y^*, **that we can not measure.**

2.1.
We assume that the observed ordinal outcome Y is a function of another variable Y*, **which is latent, continuous, and unobserved**.

2.2.
In the ordered logistic model, the latent continuous variable Y* **determines the observed ordinal categories** Y.

2.3.

The latent variable Y^* has M **threshold points** (cut-points), denoted by μ, and the observed value of Y depends on whether these thresholds are crossed.

$$Y_i = \begin{cases} 1 & \text{if } Y_i^* \leq \mu_1 \\ 2 & \text{if } \mu_1 < Y_i^* \leq \mu_2 \\ 3 & \text{if } Y_i^* > \mu_2 \end{cases}$$

Ordinal regression

Y^*: **trust in the government**

$Y^* \leq \boldsymbol{\mu}_1 = 5$ $\rightarrow Y = 1$

$(\boldsymbol{\mu}_1 = 5 < Y^* \leq \boldsymbol{\mu}_2 = 15)$ $\rightarrow Y = 2$

$(Y^* > \boldsymbol{\mu}_2)$ $\rightarrow Y = 3$

Ordinal regression

Given the continuous latent variable Y^*

These quantify the effect of each explanatory variable $X_{\{ki\}}$ in the latent variable Y_i^*.

Linear Predictor
This is the systematic part of the model, without the random component.

$$Y_i^* = \alpha + \beta^\top x_k + \varepsilon_i = Z_i + \varepsilon_i$$

Latent Continuous Variable Y^*

- Unobserved for individual i
- **It represents the underlying level of a characteristic such as satisfaction**, attitude, intention.
- Although we do not observe it directly, we assume that it determines the ordinal category that we can measure.

Explanatory Variables
These are the regressors for individual i.

- Depending on the construct, they may include age, income, or education.
- The subscript k indicates that there are K different variables.

Error Term / Random Disturbance
This captures unobserved factors or measurement errors.

- In ordered logit models, the error term is assumed to follow a standard logistic distribution with mean 0 and variance ≈ 3.29.
- In ordered probit models, it follows a standard normal distribution, N(0,1).

Ordinal regression

By denoting $\mu = (\mu_0, \mu_1, \mu_2, \ldots, \mu_j)$, **as a parameter vector with:** $\mu_0 = -\infty$ y $\mu_j = +\infty$, the **observation rule for the latent variable is given by**:

$$\begin{cases} y = 1 & \iff \mu_0 \leq \alpha + \beta^T x + \epsilon \leq \mu_1 \\ y = 2 & \iff \mu_1 \leq \alpha + \beta^T x + \epsilon \leq \mu_2 \\ \vdots & \vdots \\ y = J-1 & \iff \mu_{J-2} \leq \alpha + \beta^T x + \epsilon \leq \mu_{J-1} \\ y = J & \iff \mu_{J-1} \leq \alpha + \beta^T x + \epsilon \leq \mu_J \end{cases}$$

Taking $y = 1$, and subtracting α from both sides of the inequality we get:

$$y = 1 \iff \mu_0 \leq \alpha + \beta^T x + \epsilon \leq \mu_1$$
$$y = 1 \iff \mu_0 - \alpha \leq \beta^T x + \epsilon \leq \mu_1 - \alpha$$

Thus the observation rule does not depend on e, μ_0, μ_1 or on α, but rather on: $(\mu_0 - \alpha)$ y $(\mu_1 - \alpha)$

$$y = 1 \iff \mu_0 - \alpha - \beta^T x \leq \epsilon \leq \mu_1 - \alpha - \beta^T x$$
$$y = 1 \iff \mu_0 - \beta^T x \leq \epsilon \leq \mu_1 - \beta^T x$$

The decision regarding the value that y takes is therefore reduced to determining where the error term ε falls with respect to the thresholds adjusted by the linear predictor.

The model transforms the original thresholds into shifted thresholds, and the probability of each category is calculated as the area under the curve of $\epsilon \sim N(0,1)$ between those limits.

Then we can define F as the cumulative density function of ϵ, for a given j value of y.

$$P(y_n = j) = F(\mu_j - \beta^\top x_n) - F(\mu_{j-1} - \beta^\top x_n)$$

$$\frac{\partial \Pr[y_n = j]}{\partial x_n} = \{F'(\mu_j - \beta^\top x_n) - F'(\mu_{j-1} - \beta^\top x_n)\}$$

Ordinal regression

$$P(y_n = j) = F(\mu_j - \beta^\top x_n) - F(\mu_{j-1} - \beta^\top x_n)$$

The probability that observation *i* falls into the ordered category *j* is defined as the **difference between the cumulative distribution functions of ε, evaluated at different threshold points μ**

Cumulative distribution functions of ε, evaluated at different threshold points μ, the covariates x, and the parameters β.

The parameters β, along with the m − 1 threshold parameters $\mu_1, \dots, \mu_{\{m-1\}}$ are obtained by maximising the log-likelihood function of the model.

$$F(z) = \frac{e^z}{1+e^z}$$

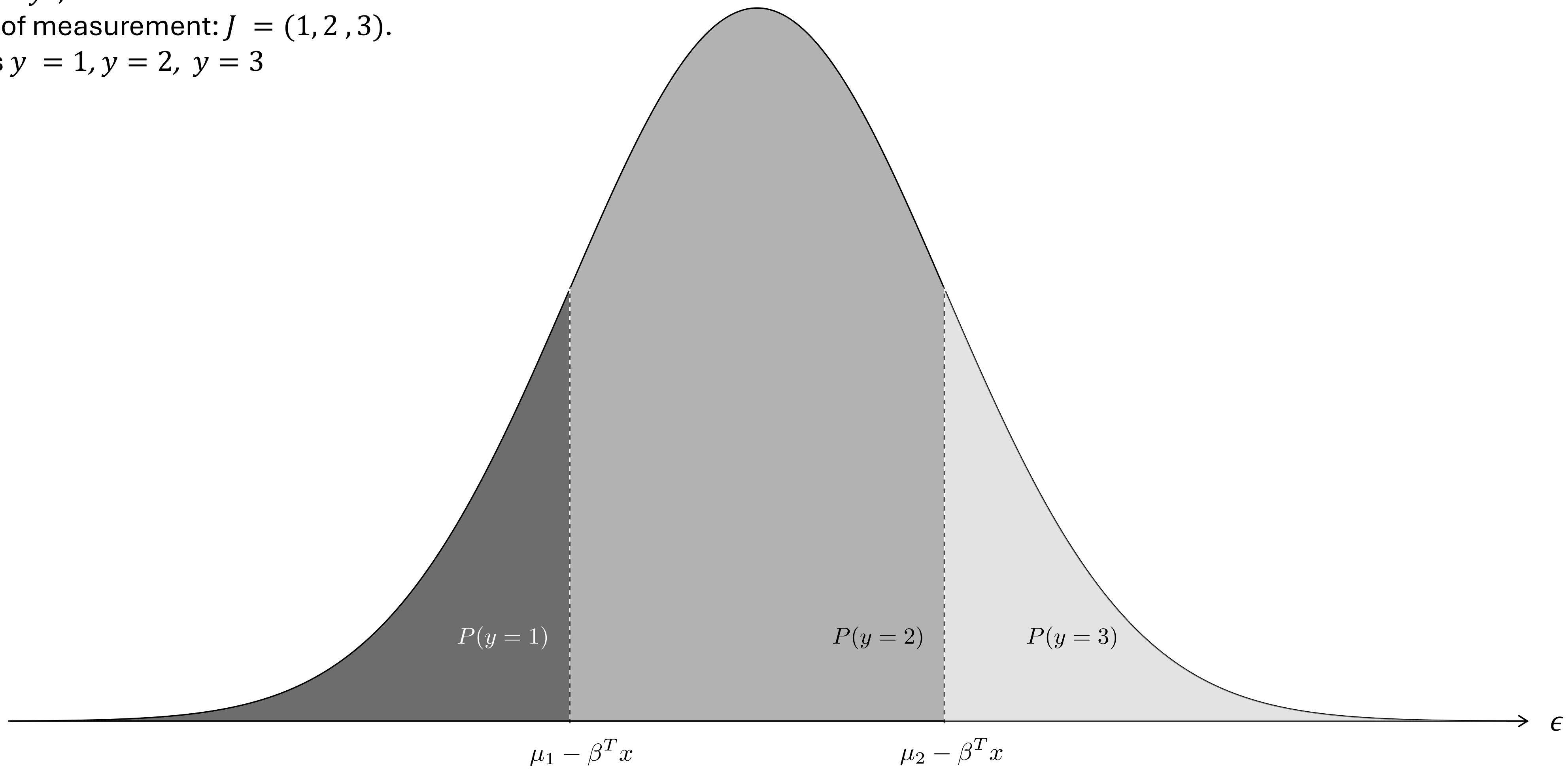
In the ordered logit model, the error term ε follows a **logistic distribution**.

$$F(z) = \Phi(z)$$

In the ordered probit model, the error term ε **follows a standard normal distribution**, and F corresponds to the CDF of that normal distribution

Ordinal regression

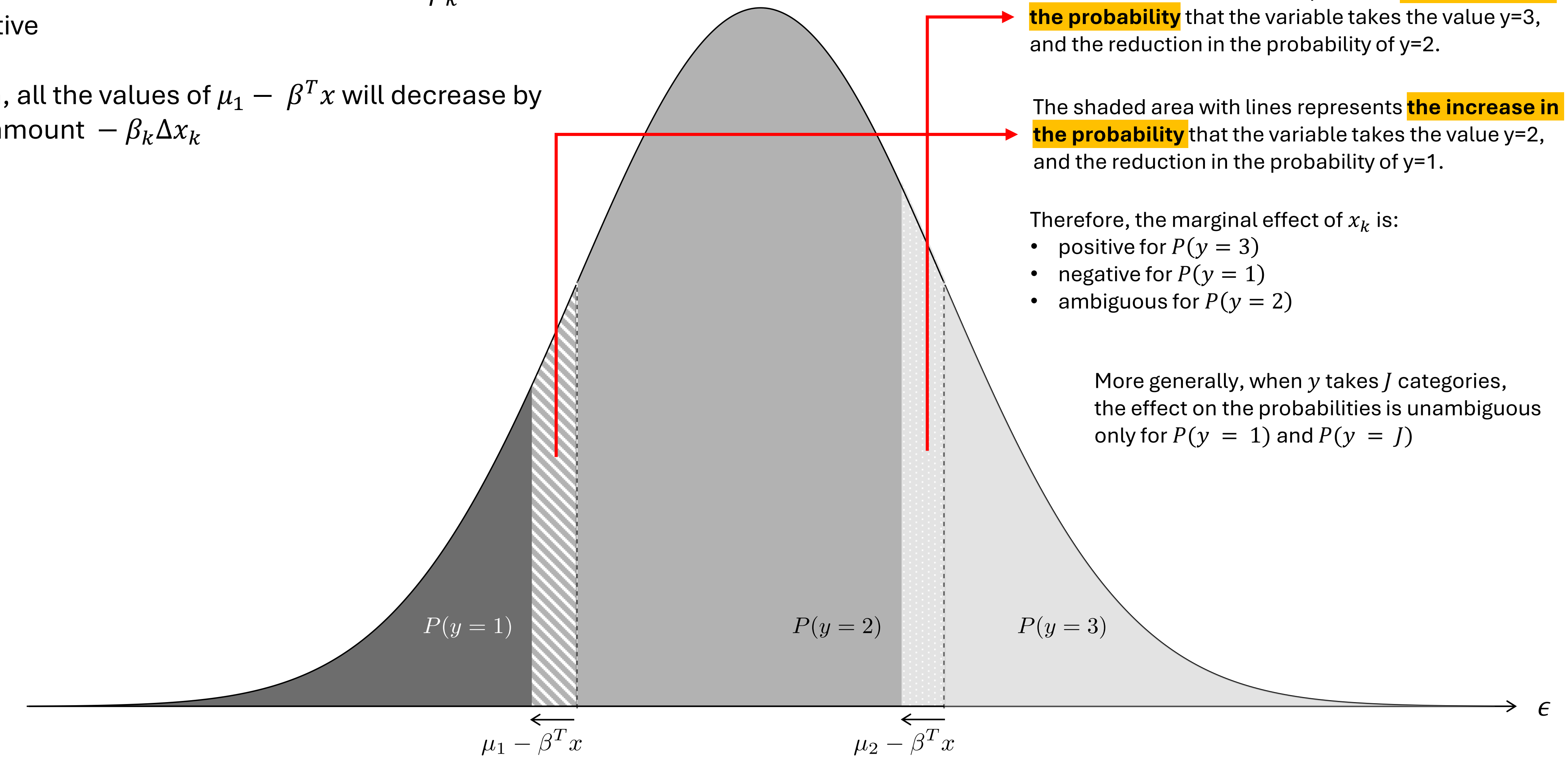
Let us suppose that for a latent variable y^*, we can observe three levels of measurement: $J = (1, 2, 3)$. That is $y = 1, y = 2, \ y = 3$



Ordinal regression

Consider now an increase in the covariate x_k, and suppose that its associated coefficient $\beta_k$is positive

Then, all the values of $\mu_1 - \beta^T x$ will decrease by the amount $-\beta_k \Delta x_k$



The shaded area with dots represents **the increase in the probability** that the variable takes the value y=3, and the reduction in the probability of y=2.

The shaded area with lines represents **the increase in the probability** that the variable takes the value y=2, and the reduction in the probability of y=1.

Therefore, the marginal effect of x_k is:

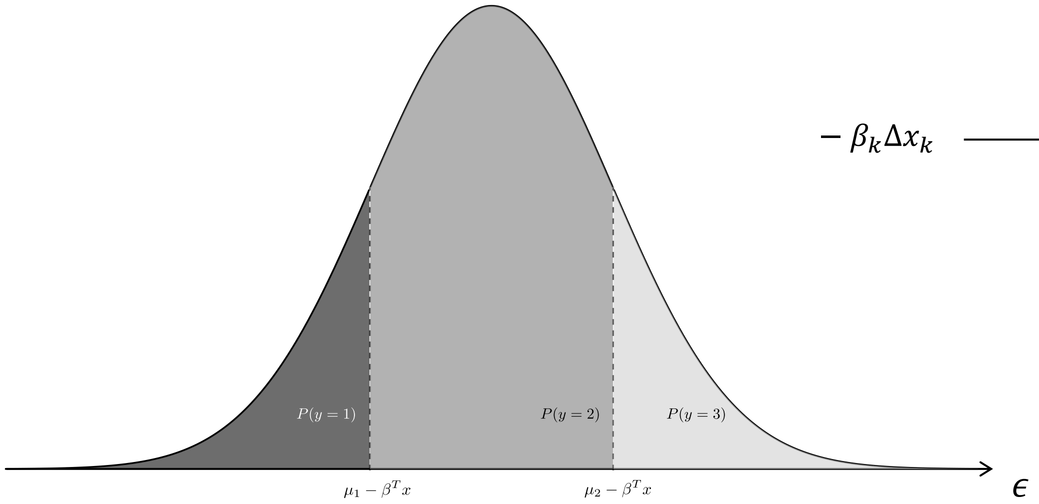
- positive for $P(y = 3)$
- negative for $P(y = 1)$
- ambiguous for $P(y = 2)$

More generally, when y takes J categories, the effect on the probabilities is unambiguous only for $P(y = 1)$ and $P(y = J)$

Ordinal regression

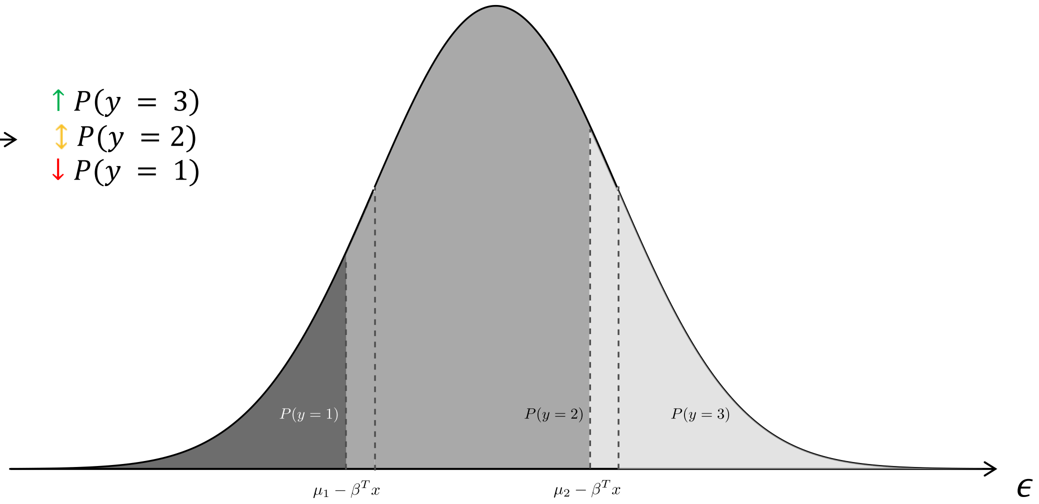
Consider now an increase in the covariate x_k, and suppose that its associated coefficient β_k is positive

Then, all the values of $\mu_1 - \beta^T x$ will decrease by the amount $-\beta_k \Delta x_k$



$-\beta_k \Delta x_k \longrightarrow$

↑ $P(y = 3)$
↕ $P(y = 2)$
↓ $P(y = 1)$



Ordinal regression

Logistic link

$$P(Y_n > j) = \frac{e^{\beta^\top x_n - \mu_j}}{1 + e^{(\beta^\top x_n - \mu_j)}}$$

The probability that observation ***i* falls into a category higher than *j***

$$P(Y_n = 1) = 1 - \frac{e^{\beta^\top x_n - \mu_1}}{1 + e^{(\beta^\top x_n - \mu_1)}}$$

Probability that observation *i* falls into category ***j* = *1*.**

$$P(Y_n = j) = \frac{e^{\beta^\top x_n - \mu_{j-1}}}{1 + e^{(\beta^\top x_n - \mu_{j-1})}} - \frac{e^{\beta^\top x_n - \mu_j}}{1 + e^{(\beta^\top x_n - \mu_j)}}$$

Probability that observation *i* falls into **category *j* .**

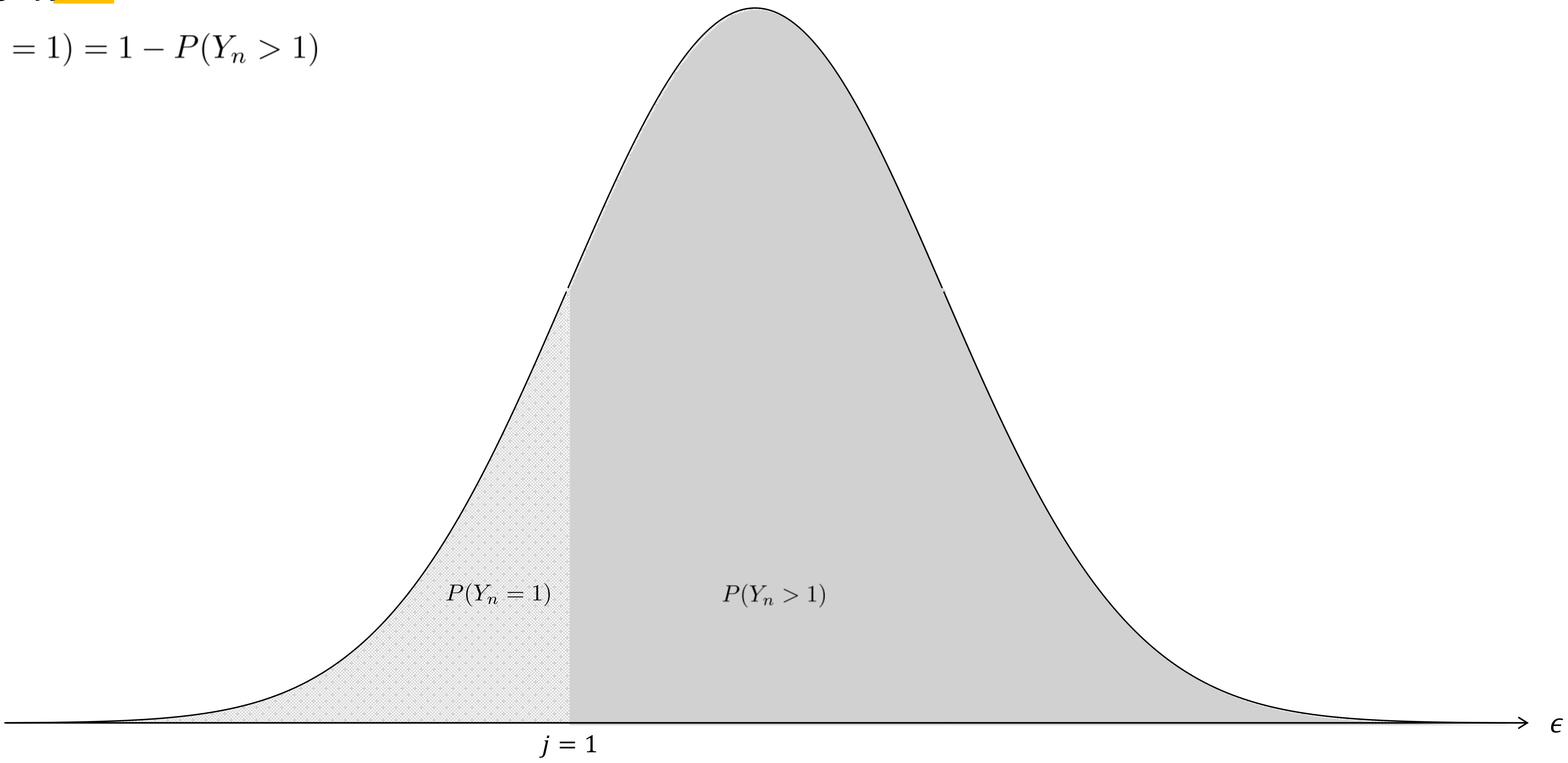
$$P(Y_n = m) = \frac{e^{\beta^\top x_n - \mu_{m-1}}}{1 + e^{(\beta^\top x_n - \mu_{m-1})}}$$

Probability that observation ***i* falls into category *m*.**

Ordinal regression

Probability that observation *i* falls into category ***j = 1.***

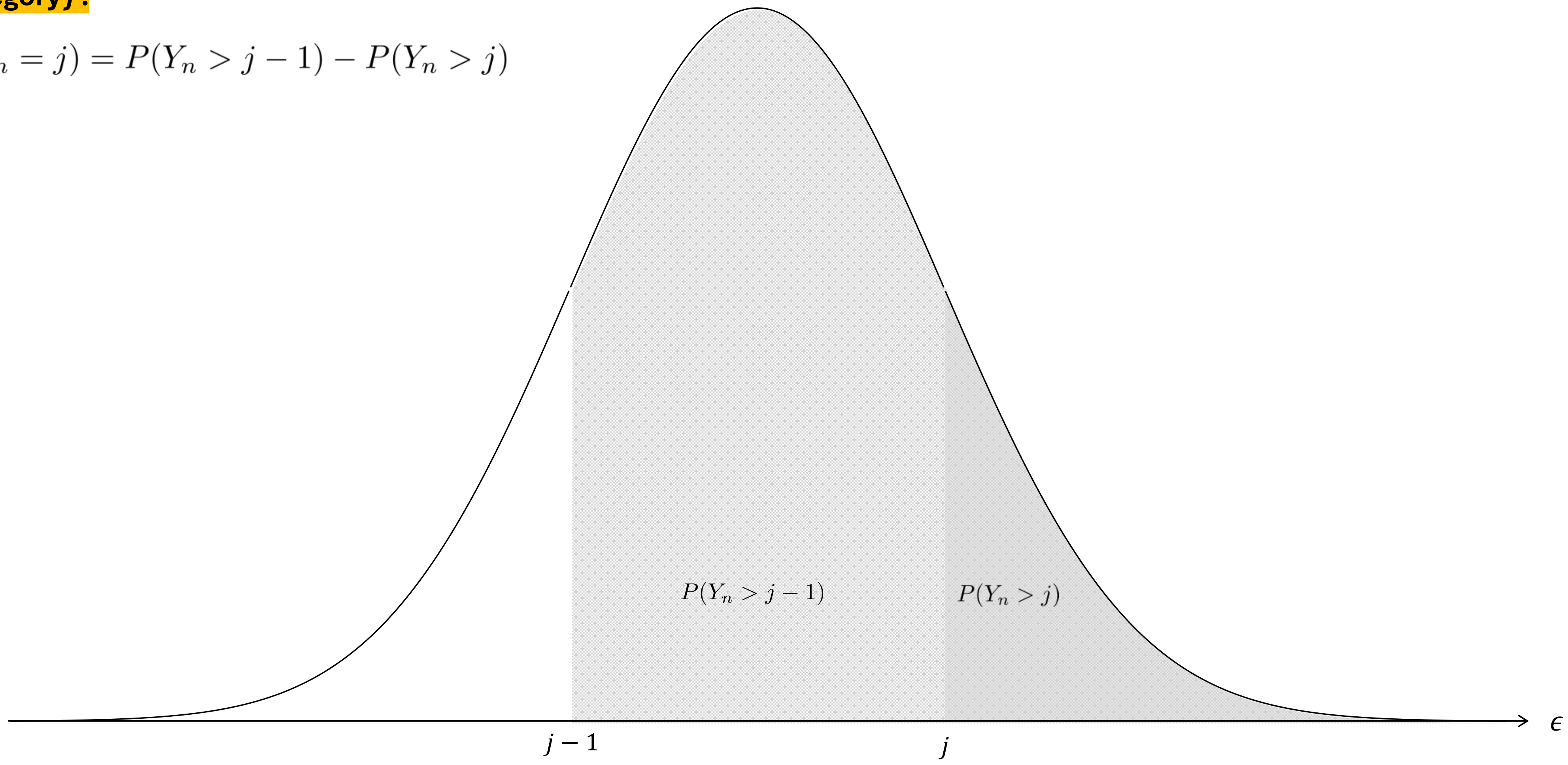
$$P(Y_n = 1) = 1 - P(Y_n > 1)$$



Ordinal regression

Probability that observation *i* falls into **category *j* .**

$$P(Y_n = j) = P(Y_n > j-1) - P(Y_n > j)$$



Ordinal regression

The specification of the ordinal logistic model is

$$P(Y_n > j) = \frac{e^{\beta^\top x_n - \mu_j}}{1 + e^{(\beta^\top x_n - \mu_j)}}$$

To obtain the odds, we need the probability of the complementary class — that is, the probability tha $Y_n$alls into category j or any lower category($Y_n \leq j$):

$$P(Y_n \leq j) = 1 - P(Y_n > j)$$

Substituting:

$$P(Y_n \leq j) = 1 - \left(\frac{e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}}{1 + e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}} \right)$$

$$P(Y_n \leq j) = \frac{1}{1 + e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}}$$

The lower cumulative odds $(Y_n \leq j)$ are defined as the ratio between:

- The probability tha Y_n is greater than *j*
- The probability that $Y_n$is less than or equal to *j*

$$\text{Odds}(Y_n \leq j) = \frac{P(Y_n \leq j)}{P(Y_n > j)}$$

Substituting:

$$\text{Odds}(Y_n \leq j) = \frac{\left(\frac{1}{1+e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}} \right)}{\left(\frac{e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}}{1+e^{\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j}} \right)}$$

$$\text{Odds}(Y_n \leq j) = \frac{1}{e^{(\boldsymbol{\beta}^\top \mathbf{x}_n - \mu_j)}}$$

By applying the natural logarithm to both sides of the equation, we obtain the log-odds:

$$\ln \left(\frac{P(Y_n \leq j)}{P(Y_n > j)} \right) = \mu_j - \boldsymbol{\beta}^\top \mathbf{x}_n$$

This is the standard form of the **Ordinal Logistic Model with Proportional Odds**, using the lower cumulative logit specification.

Ordinal regression

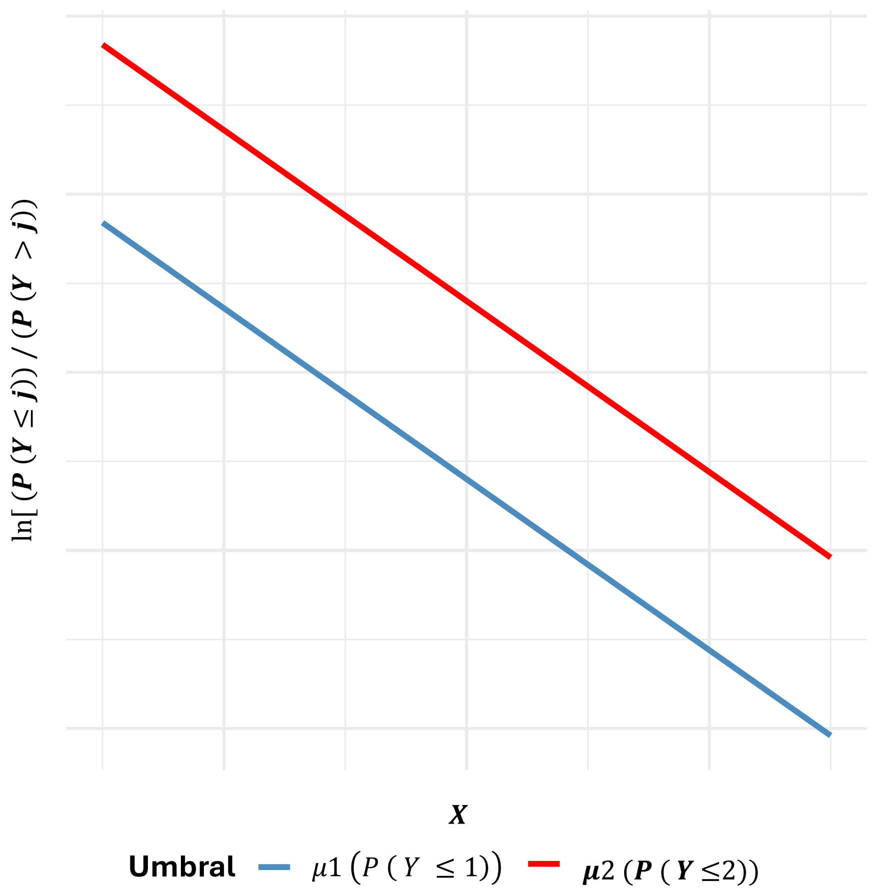
Assumption of Proportional Odds

This assumption states that the effect of any predictor variable (X) on the cumulative odds of obtaining a higher response ($Y > j$) is the same for all threshold points (μ_j) that separate the categories.

$$\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right) = \mu_j - \boldsymbol{\beta}^\top \mathbf{x}_n$$

In other words, the coefficient vector β is constant for all levels $j = 1, 2, \ldots, M-1$. he only term that varies across categories is the intercept (cut-point) μ_j.

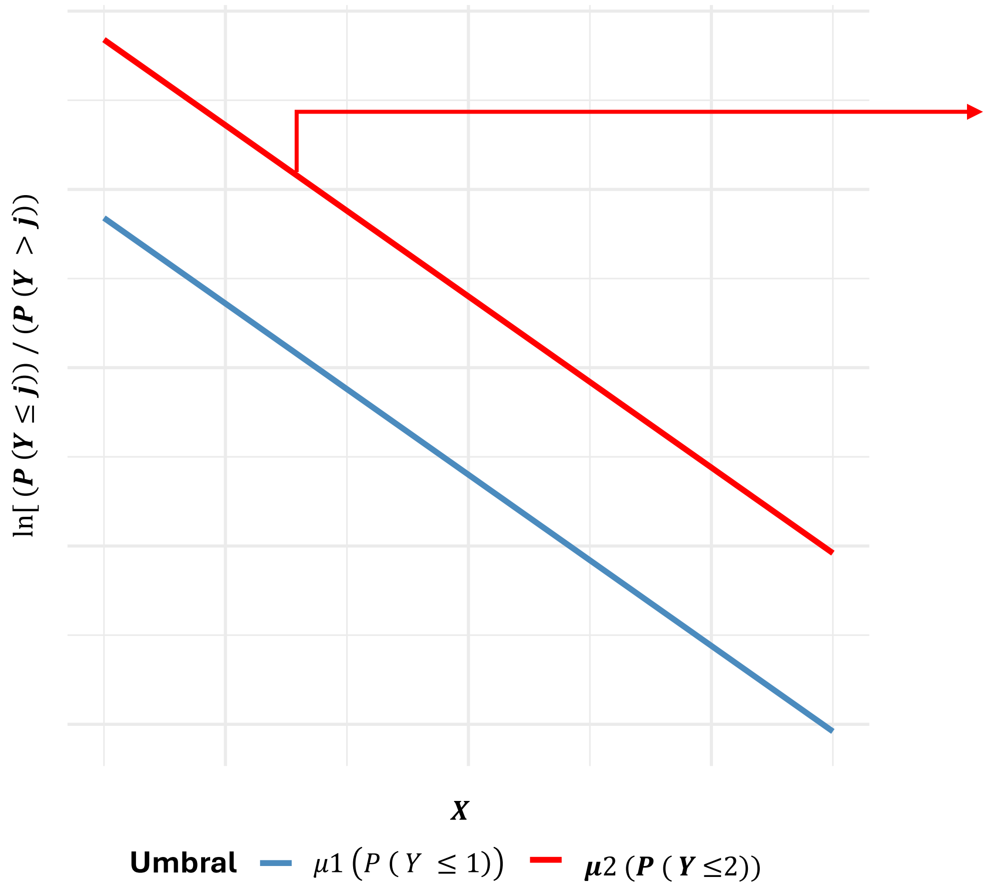
Visualizing the Proportional Odds Assumption



Ordinal regression

Assumption of Proportional Odds

Visualizing the Proportional Odds Assumption



The slope is the same for both thresholds: $-\boldsymbol{\beta} = -1.2.$
Since β is positive (1.2), the logit of $P(Y \leq j)$ is decreasing with respect to X.

An increase in the predictor X (for example, higher GPA) shifts the probabilities towards higher categories, and this happens for both thresholds shown.

As X increases, the odds ratio:

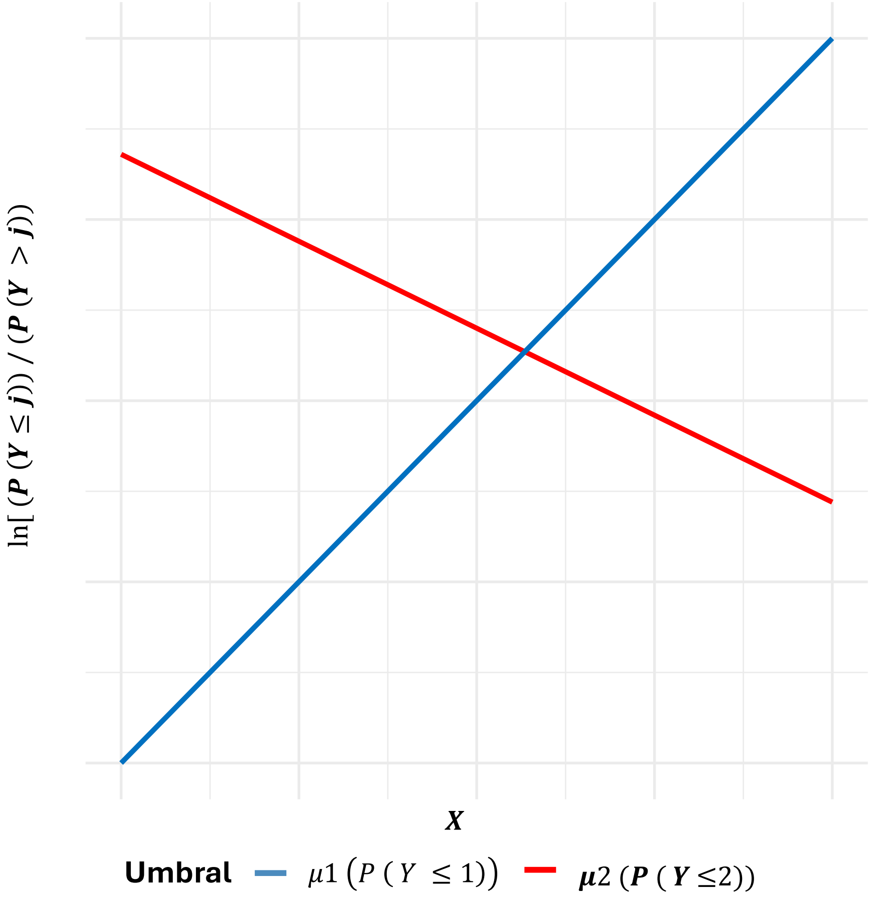
$$\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right) = \mu_j - \boldsymbol{\beta}^\top \mathbf{x}_n$$

Decreases (becoming smaller than 1), which means that the numerator ($P(Y \leq j)$) falls relative to the denominator ($P(Y > j)$):
This means that an increase in X reduces the odds of obtaining a lower category.

Ordinal regression

Assumption of Proportional Odds

Visualizing the Proportional Odds Assumption



The slope is not the same across thresholds:

- **For threshold μ2:** $-\boldsymbol{\beta} = -1.2$
- **For threshold μ1:** $-\boldsymbol{\beta} = 2.5$

- Since β for threshold 1 is positive (1.2), the logit of P(Y≤j) is decreasing with respect to X.
- However, in this case, the slope at threshold 2 is positive (+2.5), indicating that the logit of P(Y≤j) behaves differently across thresholds.

In this situation, the ordered logit model is not suitable for studying the dependent variable, because the model estimates a single coefficient β for all cut-points.

If the true slopes differ, the single estimated slope becomes an average that fits poorly at the extremes of the ordinal scale.

When this happens, one must make decisions about the model's application:

- Re-categorise the variable, if theoretically justified, and fit a new ordered logit model, or a binary logit based on the new categorisation.
- If theoretically justified, fit a multinomial logit model.
- Fit a partial proportional odds logit model, which relaxes the proportional-odds assumption.

Ordinal regression

A researcher is interested in how variables such as grade point average (GPA), the prestige of the undergraduate institution, the type of institution, and parents' education **influence the decision to apply to a postgraduate programme.**

The response variable is **ordinal**:

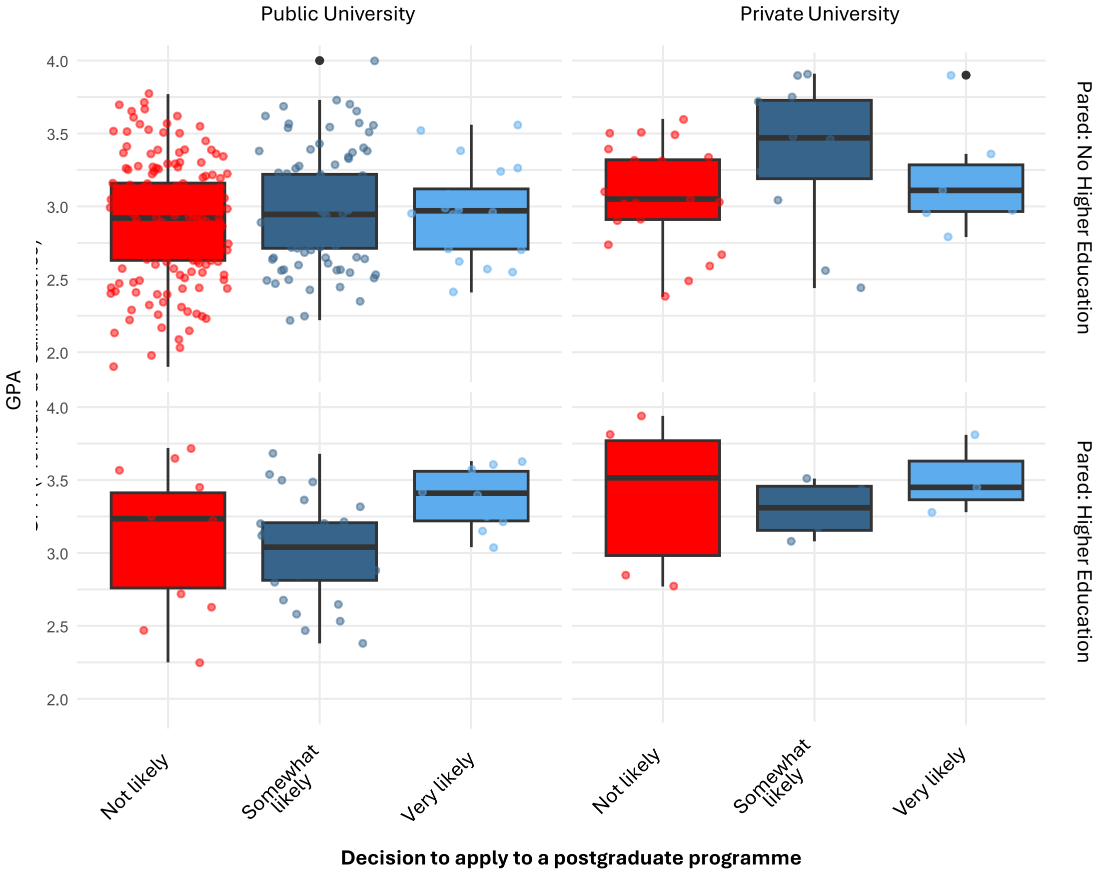
Apply: $\{not\ likely\ <\ somewhat\ likely < \ very\ likely\}$

	apply	pared	public	gpa
1	very likely	0	0	3.26
2	somewhat likely	1	0	3.21
3	unlikely	1	1	3.94
4	somewhat likely	0	0	2.81
5	somewhat likely	0	0	2.53
6	unlikely	0	1	2.59
7	somewhat likely	0	0	2.56

- **Apply.** Decision to apply to a postgraduate programme
- **Pared.** Indicator variable specifying whether the student's parents hold an undergraduate degree.
- **Public.** Indicator variable denoting whether the student completed their undergraduate studies at a public university.
- **GPA.** Independent numerical variable describing the student's cumulative academic average (Grade Point Average).

Ordinal regression

Stratified Boxplot



Relative frequencies table

Category	Level	Frequency	R frequency
A. Intention to Apply (apply)			
Not likely	1	220	55%
Somewhat likely	2	140	35%
Very likely	3	40	10%
B. Parents' Higher Education (pared)			
No Higher Education	0	337	84.25%
With Higher Education	1	63	15.75%
C. Type of University (public)			
Private	0	343	85.75%
Public	1	57	14,25%

Ordinal regression

When estimating the model, we obtain:

Regression function used

Dependent variable

Independent variables

Reference to the table containing the variables

Estimated coefficients of the ordinal logit model. Their scale is in log-odds, which means they cannot be interpreted in the same way as coefficients from an OLS regression. To interpret them correctly, researchers must:

- compute odds ratios, and
- evaluate predicted probabilities.

```
Call:
MASS::polr(formula = apply ~ pared + public + gpa, data = dat,
    Hess = TRUE)

Coefficients:
          Value Std. Error t value
pared   1.04769     0.2658  3.9418
public -0.05879     0.2979 -0.1974
gpa     0.61594     0.2606  2.3632

Intercepts:
                            Value   Std. Error t value
unlikely|somewhat likely     2.2039  0.7795     2.8272
somewhat likely|very likely  4.2994  0.8043     5.3453

Residual Deviance: 717.0249
AIC: 727.0249
```

Test statistic. Evaluates whether each coefficient differs significantly from zero.
To reject the null hypothesis associated with a coefficient, the value of the statistic must exceed the critical value corresponding to the chosen significance level α (1.96 when α = 0.05).

Null deviance (model without predictors). Indicates how well (or poorly) the model predicts the response variable when no control variables are included. Higher values imply a poorer fit.

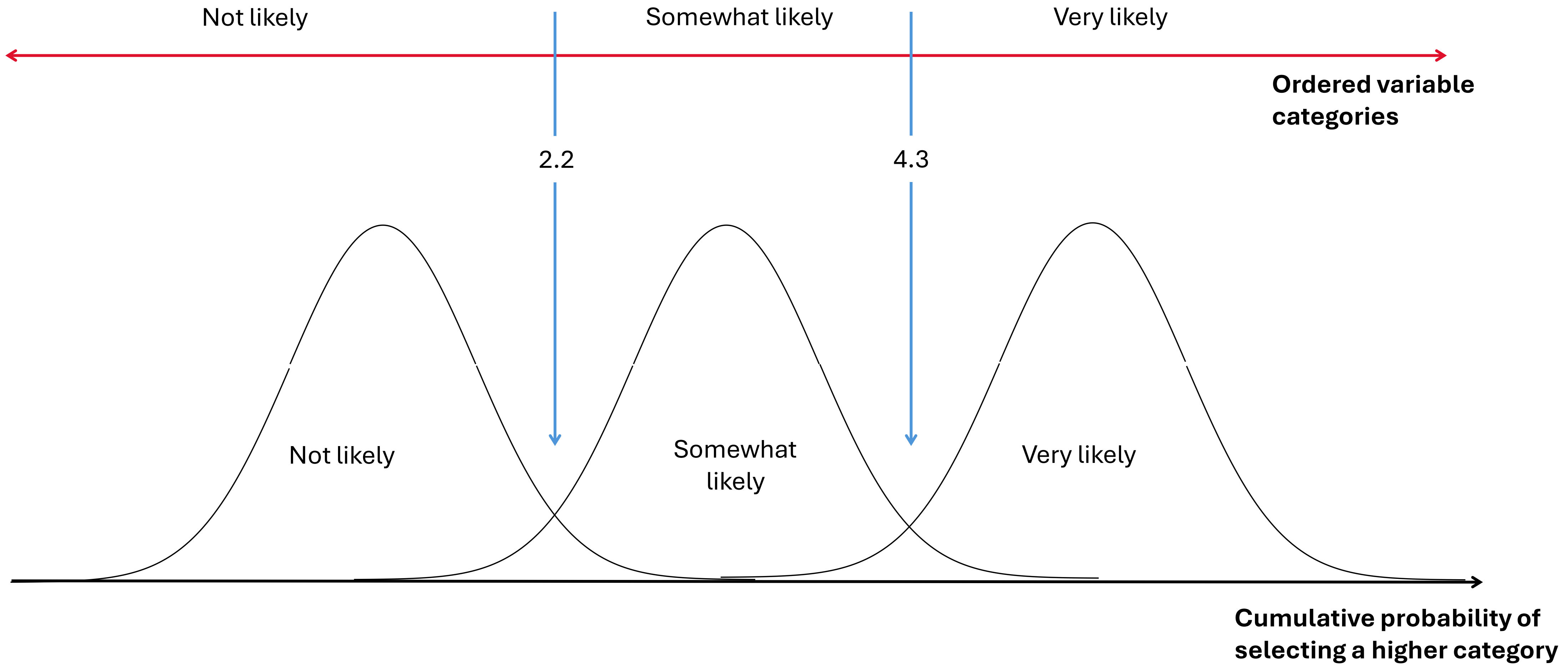
Intercepts (cut-points) of the model. These are not direct probabilities. Instead, they correspond to the threshold parameters (μ_k) on the latent, underlying continuous scale (Y*), expressed on the logit scale.

These cut-points determine where, along the linear predictor, **the cumulative probabilities transition from one category to the next.**

Akaike Information Criterion (AIC). Penalises model complexity. A lower AIC reflects a better balance between goodness-of-fit and parsimony.

Ordinal regression

Estimated Intercepts (Cut-points)

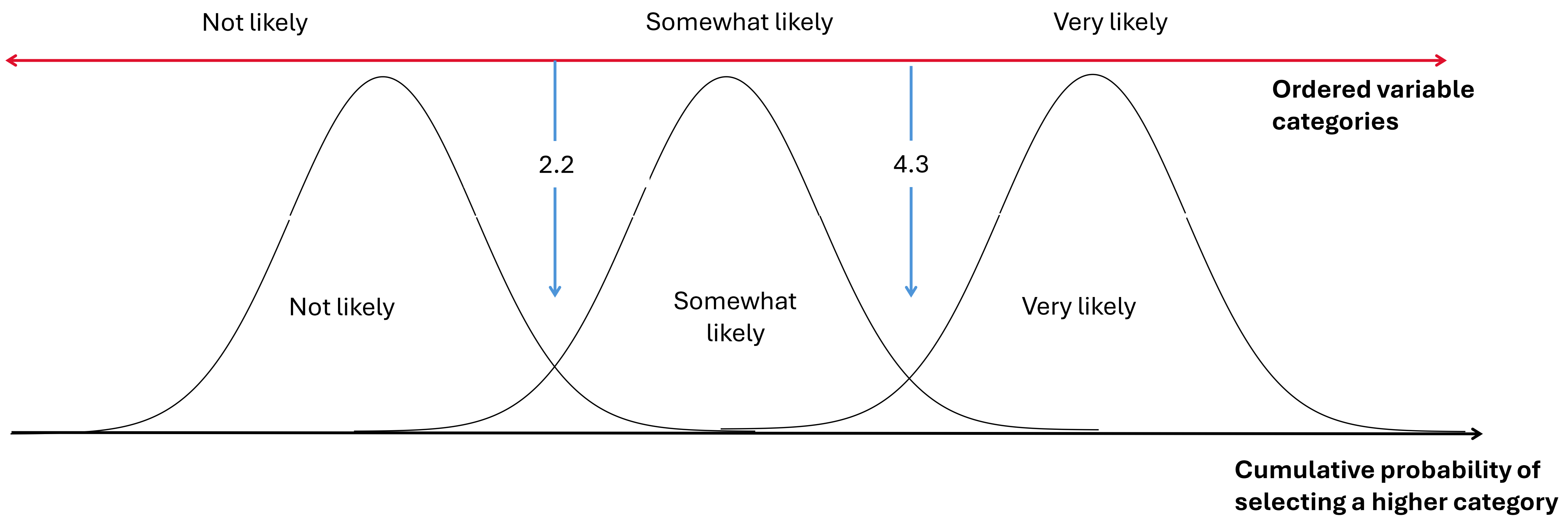


Ordinal regression

Estimated Intercepts (Cut-points)

The estimated cut-points at 2.2 and 4.3 determine the boundaries separating “not likely” from “somewhat likely”, and “somewhat likely” from “very likely”, respectively.

The categories of the ordinal response variable correspond to intervals on the underlying latent scale. These latent thresholds do not represent direct probabilities; instead, they indicate the points at **which the cumulative probability of selecting a higher category crosses specific values**.



Ordinal regression

Results of the Ordinal Logistic Regression on the Intention to Apply to a Postgraduate Programme

Variable	Coefficient	Standard Error	t-Statistic	
pared	1.048	0.266	3.942	$p < 0.001^{***}$
public	-0.059	0.298	-0.197	$p = 0.844$
gpa	0.616	0.261	2.363	$p = 0.018^{*}$
Category Thresholds				
unlikely | somewhat likely	2.204	0.780	2.827	$p = 0.005^{**}$
somewhat likely | very likely	4.299	0.804	5.345	$p < 0.001^{***}$

Significance codes: $^{***}p < 0.001$ $^{**}p < 0.01$ $^{*}p < 0.05$
Residual deviance: 717.02
AIC: 727.02

$$\ln\left(\frac{P(Y_n \le j)}{P(Y_n > j)}\right) = \mu_j - \boldsymbol{\beta}^\top \mathbf{x}_n$$

- **Having parents with higher education.** This factor reduces the cumulative log-odds of being in a lower application-intention category compared with being in the highest category ("very likely") by 1.048 units. **In substantive terms, students whose parents hold a university degree are more likely to report a higher intention to apply.**

- **Graduating from a public university.** Shows a coefficient that is slightly positive (0.059) but statistically insignificant. Therefore, there is no evidence that attending a public university affects the intention to apply to a postgraduate programme.

- **Grade Point Average (GPA).** Each additional point in GPA decreases the cumulative log-odds of being in a lower category by 0.616 units, indicating that higher academic achievement is associated with higher intention to apply.

The estimated model includes threshold (cut-point) parameters and cumulative probabilities for the lower-order categories. These cut-points indicate the values of the latent variable at which the probability of selecting a higher category exceeds that of the adjacent lower category.

$$\ln\left(\frac{\hat{P}(Y \le 1)}{\hat{P}(Y_n > 1)}\right) = 2.204 - (1.048 \cdot \text{pared} + -0.059 \cdot \text{public} - 0.616 \cdot \text{gpa})$$

$$\ln\left(\frac{\hat{P}(Y \le 2)}{\hat{P}(Y_n > 2)}\right) = 4.299 - (1.048 \cdot \text{pared} + -0.059 \cdot \text{public} - 0.616 \cdot \text{gpa})$$

To aid interpretation, the exponentiated coefficients (odds ratios) should be computed .These measure the multiplicative change in the cumulative odds of reporting a higher intention to apply, associated with a one-unit increase in each predictor.

$$e^{\ln\left(\frac{P(Y_n \le j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} = \frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \le j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} = \frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Parental education

For students, whose parents completed higher education, **the probabilities of being in a higher category of intention to apply** (that is, “somewhat likely” or “very likely”, compared with “not likely”) **are 2.85 times greater than for students whose parents do not hold a university degree, holding all other variables constant.**

This corresponds to an increase of approximately 185%:

$$(2.8511 - 1) \times 100 \approx 185\%$$

Using the lower cumulative odds (Logit $de\ P(Y \le j)$ the inverse odds ratio is:

$$\mathrm{OR}_{\mathrm{Inferior}} = 1/\mathrm{OR}$$

$$\mathrm{OR}_{\mathrm{Inferior}} = \frac{1}{2.85} \approx \mathbf{0.35}$$

Thus, the lower cumulative odds (“not likely” vs “somewhat or very likely”) are 65% lower for students whose parents have higher education:

$$(1 - 0.35) \times 100 \approx 65\%$$

For students whose parents did attend university**, the probabilities of being in a lower category of intention to apply** are 65% lower than for students whose parents do not have higher education.

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \leq j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} = \frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \leq j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

Type of University

For students graduating from a public university, **the probabilities of being in a higher intention-to-apply category** (that is, "somewhat likely" or "very likely", as opposed to "not likely") are 0.9429 times those of students graduating from private universities, holding all other variables constant.

This corresponds to a 5.71% decrease:

$$(1 - 0.9429) \times 100 \approx 5.71\%$$

Using the lower cumulative odds (Logit $de\ P(Y \leq j)$ the inverse odds ratio is:

$$\mathrm{OR}_{\mathrm{Inferior}} = 1/\mathrm{OR}$$

$$\mathrm{OR}_{\mathrm{Inferior}} = \frac{1}{0.9429} \approx \mathbf{1.06}$$

Thus, the lower cumulative odds ("not likely" vs. "somewhat or very likely") are 6% higher for students graduating from public universities.

$$(1.06 - 1) \times 100 \approx 6\%$$

For students who graduated from a public university**, the probabilities of being in a lower category of intention to apply** are 6% higher than those of students who graduated from private universities.

Ordinal regression

Results of the Ordinal Logistic Regression Regarding the Intention to Apply to Postgraduate Studies

Variable	OR	Lower limit	Upper limit
pared	2.8511	1.6958	4.8171
public	0.9429	0.5209	1.6806
gpa	1.8514	1.1136	3.0985

Note: 95% confidence intervals for the odds ratios.

To interpret the results, we may (and should) also apply the exponential function to the estimated parameters to obtain the odds ratios (ORs):

$$e^{\ln\left(\frac{P(Y_n \le j)}{P(Y_n > j)}\right)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \quad = \frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})}$$

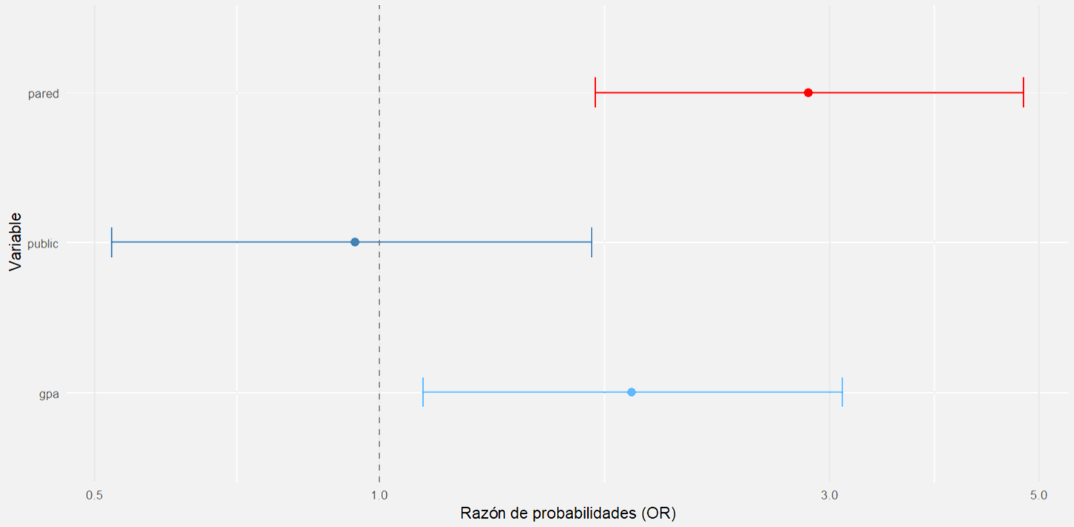
$$\frac{P(Y_n \le j)}{P(Y_n > j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \longrightarrow e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x} + \beta_j)} = e^{(\mu_j - \boldsymbol{\beta}^\top \mathbf{x})} \cdot e^{(\beta_j)}$$

GPA

For each additional unit of GPA, the **probabilities of being in a higher intention-to-apply category** increase by a factor of 1.85, which corresponds to an 85% increase, holding all other variables constant.

$$(1.8514 - 1) \times 100 \approx 85\%$$

Forest Plot of the Estimated Parameters in the Ordinal Model



Ordinal regression

Proportional Odds assumption

To verify whether the effect of any predictor variable (X) on the cumulative odds of obtaining a higher response (Y > j) is the same across all threshold points (μ_1, μ_2, **...), we estimate auxiliary binary logit models based on dichotomisations of the ordinal dependent variable:**

1 $$\frac{p}{1-p} = e^{(\boldsymbol{\beta}^\top \mathbf{x})} \qquad Y > 1 = \begin{cases} 1 & \text{if } y = \text{"somewhat likely" or "very likely"} \\ 0 & \text{if } y = \text{"not likely"} \end{cases}$$

2 $$\frac{p}{1-p} = e^{(\boldsymbol{\beta}^\top \mathbf{x})} \qquad Y > 2 = \begin{cases} 1 & \text{if } y = \text{"very likely"} \\ 0 & \text{if } y = \text{"not likely" or "somewhat likely"} \end{cases}$$

Auxiliary logit models

Model	Variable	OR	Lower limit	Upper limit	p-value
$y > 1$	Intercept	0.138	0.028	0.676	0.0146
$y > 1$	pared	2.89	1.61	5.17	$< 0.001^{***}$
$y > 1$	public	0.818	0.450	1.490	0.5113
$y > 1$	gpa	1.73	1.01	2.95	0.0442^{*}
$y > 2$	Intercept	0.0086	0.0006	0.134	$< 0.001^{***}$
$y > 2$	pared	2.50	1.18	5.29	0.0167^{*}
$y > 2$	public	1.71	0.749	3.89	0.2034
$y > 2$	gpa	2.09	0.853	5.11	0.1071

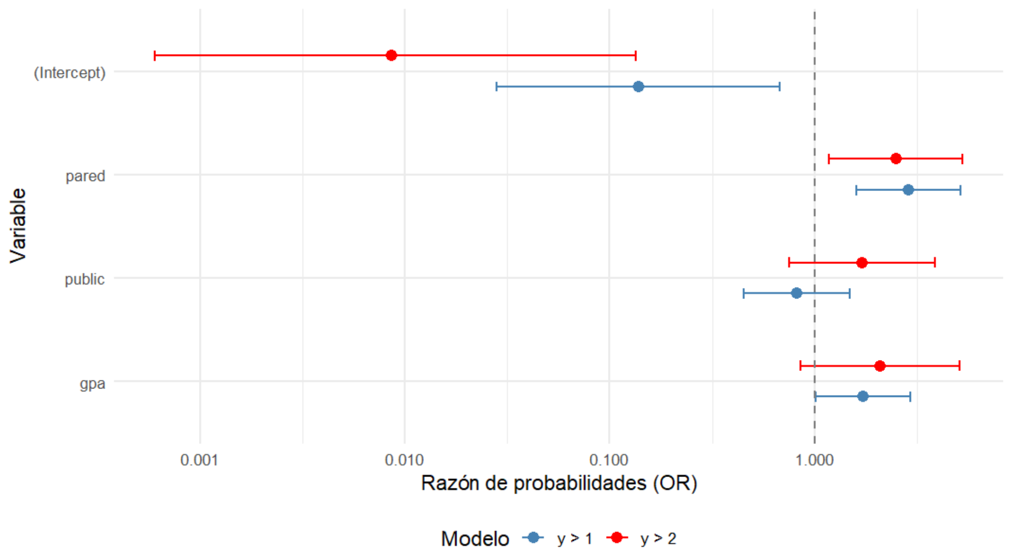
Significance codes: $^{***}p < 0.001$ $^{**}p < 0.01$ $^{*}p < 0.05$

Note: Each logit model is estimated on a different dichotomisation of the ordinal variable apply.

Model $y > 1$: compares "somewhat likely" and "very likely" against "not likely".

Model $y > 2$: compares "very likely" against "somewhat likely" and "not likely".

Forest Plot of the Estimated Parameters in the Auxiliary Logit Models



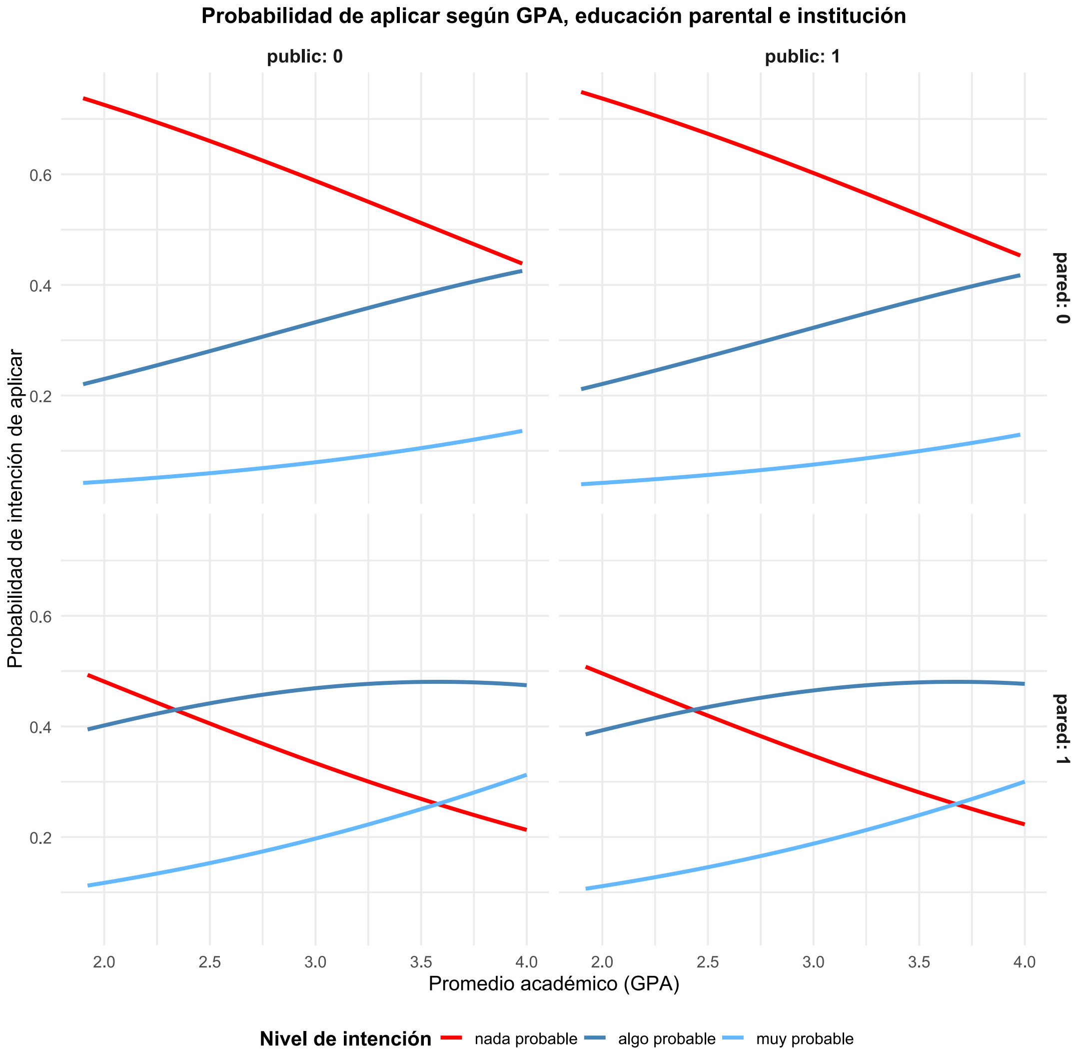
As an additional step, a Brant test may be used to evaluate formally whether the proportional-odds (parallel-regression) assumption is satisfied. The test compares the coefficients estimated in the separate binary logit models for each threshold (e.g., Y>1 and Y>2).

Test	Chi-square (X^2)	df	p-value
Omnibus	4.34	3	0.23
pared	0.13	1	0.72
public	3.44	1	0.06
gpa	0.18	1	0.67

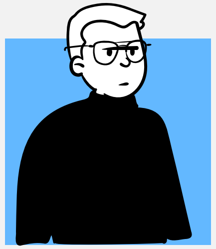
Null hypothesis (H0): The proportional-odds (parallel regression) assumption holds.

Interpretation: High p-values indicate insufficient evidence to reject H0.

Ordinal regression



The logistic model not only enables the estimation of individual probabilities but also allows the construction of comparative intention-to-apply profiles. By holding GPA constant and systematically varying parental education and institution type, it becomes possible to identify specific probability scenarios for belonging to the highest category of intention to apply.



Sebastián

Type of institution: private
Parents: both hold postgraduate degrees
GPA: 3.5

Predicted probability of belonging to the highest intention-to-apply category: 73%

Felipe

Type of institution: public
Parents: neither parent holds an undergraduate degree
GPA: 3.5

Predicted probability of belonging to the highest intention-to-apply category: 47%